

Samuel Charry

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TECHNICAL SKILLS

Programming languages: Python, SQL, JavaScript, Java, Dart

High-performance computing: Parallel processing, distributed computing

Software development: Flutter, Firebase, Figma, offline-first architectures

Machine Learning: PyTorch, Hugging Face Transformers (XLM-RoBERTa, EuroBERT), scikit-learn, TensorFlow Lite Micro, Flower (Federated Learning)

Languages: Spanish (native), English (IELTS 7.5), Portuguese (basic)

EDUCATION

Universidad de los Andes, Colombia Jan 2023 – Mar 2027 (in progress)

Systems and Computer Engineering

PROJECTS

Federated Learning on Microcontrollers – Undergraduate Thesis 2026 – Present

- Evaluating the feasibility of Federated Learning on extremely resource-constrained microcontrollers (ESP32, STM32), an area with a documented empirical gap (0 arXiv results combining FL with ESP32/STM32).
- Comparing four strategies – centralized training, FedAvg, FedAvg with gradient quantization, and FedProx – using Flower and TensorFlow Lite Micro.
- Measuring memory, energy, bandwidth, latency, and accuracy on real hardware, under the direction of Professor Lozano.

ENAd Móvil – Mobile Development (ISIS-3510) 2025 – Present

- Led the design of an offline-first mobile application to digitize the ENAd (Evaluación del Nivel Adecuado) methodology, enabling rural teachers to assess students by learning level and form dynamic groups without internet connectivity, in collaboration with the Aprender a Quererte foundation.
- Conducted user research (interviews with foundation staff and rural educators) and facilitated a Diverge-Converge process over 42 ideas, evaluated with a weighted 5-criteria matrix.
- Defined 10 Business Questions (types 1–4) to measure teacher adoption and learning outcomes; designed an offline-first architecture (SQLite + Firebase sync) and Figma prototypes; worked in a 6-person agile team.
- **Stack:** Flutter/Dart, Firebase, Figma.

Multiclass Text Classification – Kaggle Competition (Data Science Course) 2026

- Placed 5th (out of 30+ teams) on the private Kaggle leaderboard in a competition classifying historical Spanish-language documents across 39 classes, with a final score of 0.34343.
- Designed an ensemble combining XLM-RoBERTa Large and EuroBERT-210m with SVM over character n-gram TF-IDF; empirically found that raw text outperformed preprocessed text.
- Developed skills in hyperparameter tuning, cross-validation, handling class imbalance, and running deep learning pipelines under compute constraints (Kaggle GPU T4/P100).

EXPERIENCE

Universidad de los Andes – HPC Support 2024 – Present

- Provide technical support to users of the university's high-performance computing infrastructure, assisting with cluster job execution and software environment management.
- Diagnose and resolve configuration, dependency, and performance issues in parallel and distributed computing environments.
- Collaborate with researchers and students to adapt their workflows to the cluster architecture.

CERTIFICATIONS

- Introduction to High-Performance Computing (edX)
- IELTS Academic – Overall Band Score: 7.5, C1, 2026